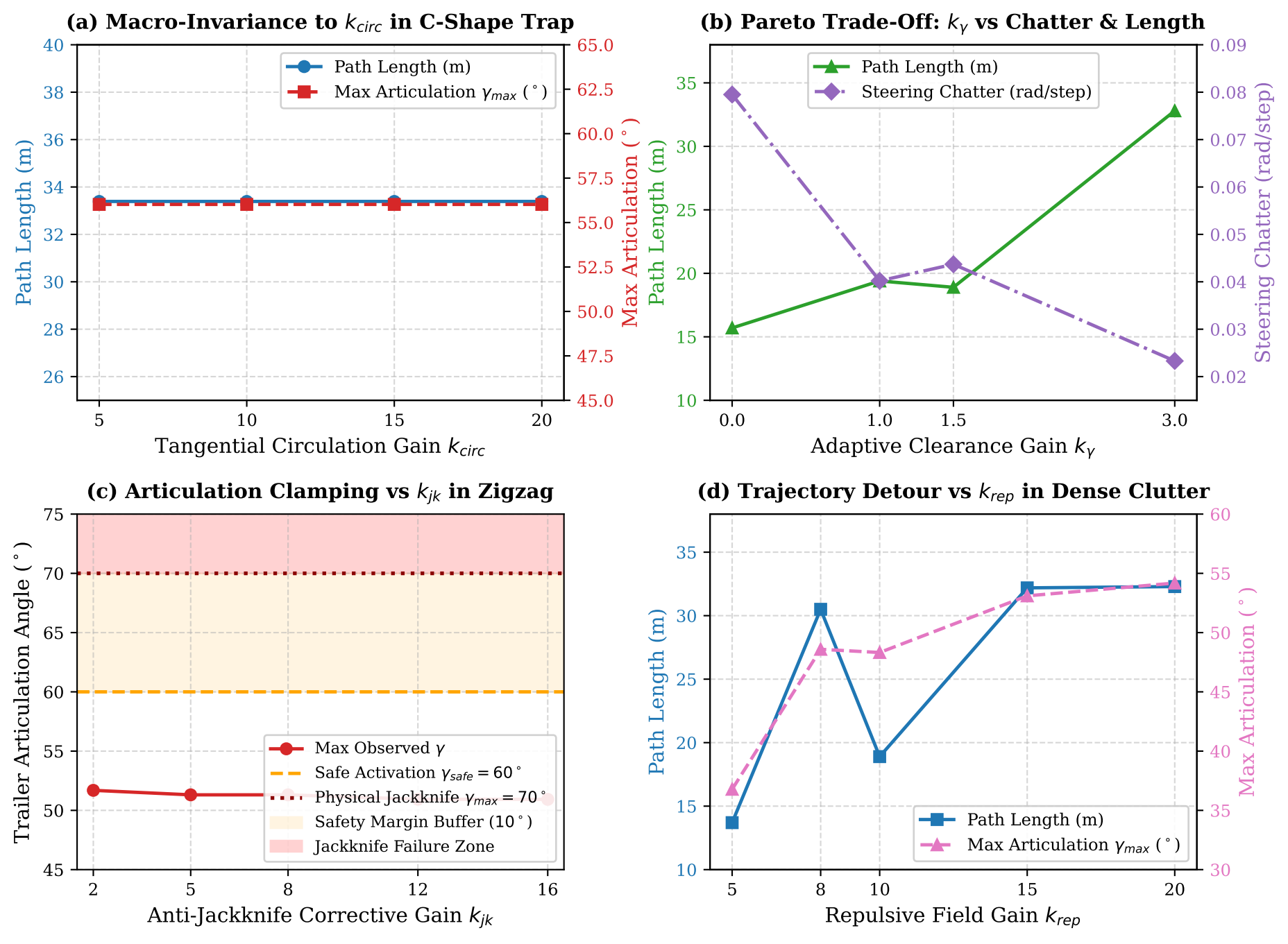
(a) Macro-Invariance to k_{circ} in C-Shape Trap
Path Length (m)
Max Articulation γ_{max} (°)
Tangential Circulation Gain k_{circ}
(b) Pareto Trade-Off: k_γ vs Chatter & Length
Path Length (m)
Steering Chatter (rad/step)
Adaptive Clearance Gain k_γ
(c) Articulation Clamping vs k_{jk} in Zigzag
Trailer Articulation Angle (°)
Max Observed γ
Safe Activation $\gamma_{safe} = 60°$
Physical Jackknife $\gamma_{max} = 70°$
Safety Margin Buffer (10°)
Jackknife Failure Zone
Anti-Jackknife Corrective Gain k_{jk}
(d) Trajectory Detour vs k_{rep} in Dense Clutter
Path Length (m)
Max Articulation γ_{max} (°)
Repulsive Field Gain k_{rep}